

# Identifying Drivers of Internal Displacement in Mali

## A Panel Regression Analysis with Random Forest Validation

### Overview

We model monthly changes in log-IDP counts across Mali's 50 admin2 cercles (2014–2024) as a function of three candidate driver families: **conflict**, **climate**, and **food prices**. Our goal is to identify which drivers causally matter, in what form, and at what lag — while being honest about what a panel regression can and cannot establish.

The analysis proceeds in two parts:

1. **Model selection** (Stages A–E): a structured funnel that narrows 282 candidate features to the 11 retained in the final model (M5), growing the specification from 1 feature to the final model one step at a time.
2. **Validation and extensions** (Parts II–IV): Random Forest stress-tests the linear model, heterogeneity tests reveal what the average effects hide, and the independent refugee-outflow outcome cross-validates the findings.

**Data.** IDP source: DTM survey panel, zero-run-filtered and re-stitchedSample: n = 1,417 observations, 44 cercles, after requiring complete data for all M5 features. Conflict: ACLED strict (direct lethal violence). Climate: ERA5 reanalysis. Food: WFP price monitoring. Spatial weights: road-distance decay matrix.

**Survey structure — irregular rounds and zero-change intervals.** DTM runs 81 irregular survey rounds (2014–2024) with non-uniform gaps, and many cercles miss rounds; in the raw panel ~65% of adjacent-round changes are **exactly zero** (repeated figures, not fresh surveys). We apply a **months-based zero-run filter**: a run of consecutive zero-change rounds is kept when the non-zero changes bracketing it are **≤5 months apart**, but a longer run is dropped and the series re-stitched into a single survivor-to-survivor interval (3,078 → 1,622 intervals, zeros 65% → 33%, gaps now 1–96 months). The outcome is then expressed **per month** (÷**months\_between**) so the re-stitched intervals stay comparable, and intervals longer than 6 months are excluded — leaving **n = 1,417** across 44 cercles.

#### Notation.

- $y_{it}$  = monthly change in log-IDP (**y\_change\_per\_month**) for cercle  $i$  at survey date  $t$
- $C_{it}$  = controls: **log1p\_idp\_t1** (AR anchor), **months\_between**
- $\alpha_i, \gamma_t$  = cercle and time fixed effects; SE clustered on cercle
- $f(\cdot)$  = feature transform: **log1p** for non-negative counts, identity for signed anomalies
- $Wx$  = spatial-spillover twin of  $x$  (neighbour-weighted average at the same lag band)

**Three criteria for variable inclusion** (applied throughout):

1. **ΔR²\_within** — does the variable add explanatory power?
2. **p-value** — is the signal distinguishable from noise?
3. **Coefficient sign** — is the direction mechanistically coherent?

No variable earns its place on one criterion alone.

### Feature construction

**Outcome.** The regression target is **y\_change\_per\_month** = (log1p(IDP\_t2) – log1p(IDP\_t1)) / **months\_between** — the average monthly log-IDP growth over each survey interval. Dividing by **months\_between** makes observations comparable regardless of survey gap length (intervals range from 1 to 6 months).

**Transforms.** The transform for each feature is chosen by value distribution:

- **Conflict** (**fatalities**, **events**): always non-negative counts → **log1p** transform. This compresses the heavy right tail (rare extreme events) and makes the relationship approximately linear on a proportional scale.
- **Climate**: all 112 climate local features use **identity (no transform)**. Most are signed anomalies (SPI, temperature, rainfall, streamflow, NDVI, soil moisture) that can be negative, so log1p is not applicable. The SPI-derived duration and severity variables (**spi6\_flood\_duration**, **spi6\_drought\_severity**, etc.) are non-negative, but are kept on identity via an explicit override — these are already-derived intensity measures where log1p would distort the natural scale and reduce |t| empirically.
- **Food**: **inflation\_food\_price\_index** (a growth rate, signed) → identity. **c\_food\_price\_index** and **c\_maize** (price levels, non-negative) → **log1p**.

**Lag structure — banded windows.** Rather than point lags (the value at exactly N months ago), we use **non-overlapping banded window averages**. A band **[lo, hi]m** takes the mean of the monthly variable over the window from **month\_of(t1) – hi** to **month\_of(t1) – lo**, strictly before the observation start **t1**. For example:

| Band   | window                      | interpretation        |
|--------|-----------------------------|-----------------------|
| 1–3m   | months 1, 2, 3 before t1    | short-term pre-survey |
| 7–9m   | months 7, 8, 9 before t1    | medium-term           |
| 10–12m | months 10, 11, 12 before t1 | ~one year lag         |
| 13–18m | months 13–18 before t1      | slow-burn effects     |
| 19–24m | months 19–24 before t1      | long-term             |

Banded windows are **non-overlapping** — the 1–3m and 4–6m bands contain different months, so they can be included jointly without multicollinearity from overlap. This is why distributed-lag models (including all bands simultaneously) remain feasible up to the point where between-band correlations become a problem.

Conflict uses bands 1–3m, 4–6m, 7–9m, 10–12m (no longer lags — conflict is not expected to have displacement effects at 2+ year horizons). Climate and food use the same four short bands plus 13–18m and 19–24m (slow-burn pathways exist for climate). The **current window** (0 months lag, i.e. the observation period **(t1, t2]** itself) is used only for climate and food — conflict in the current window is excluded because it risks reverse causality (displacement events may *cause* conflict rather than vice versa).

**Matching monthly predictors to the irregular intervals.** Conflict, climate, and food are monthly series, while the IDP intervals are irregular and — after re-stitching — of varying length. They are aligned by calendar month, by **anchoring every lag to the interval start t1**:

- A **lagged band [lo, hi]m** is the mean of the monthly series over the calendar months `month(t1)-hi ... month(t1)-lo`, strictly before **t1**. Because it depends on **t1** alone (not on **t2** or the gap length), every interval — short or long, re-stitched or not — receives the same well-defined pre-interval history. Irregular gaps therefore do not distort the lagged predictors.
- The **current window** (lag 0, climate and food only) is the mean over the interval itself, **(t1, t2]**, so it automatically spans the actual gap — averaging one month for a 1-month interval and, say, six for a re-stitched 6-month one. Pairing this with the per-month outcome keeps long and short intervals comparable.
- **Conflict has no current window** (excluded above), so it enters only through pre-**t1** bands and is wholly unaffected by interval length.

So a re-stitched long interval changes only its *own*-window average (and the outcome, which is per-month normalised); the conflict and lagged-climate history that does most of the explanatory work is anchored to **t1** and unchanged.

**Cumulative windows.** Cumulative features (e.g. total fatalities over the past 12 months) were considered in early exploratory work but **not used in the final analysis**. They overlap with adjacent banded windows, creating multicollinearity, and the banded design already captures sustained exposure via multiple non-overlapping bands. The final feature set contains banded windows only.

**Feature count.** The full candidate set has **282 features**: 16 conflict (2 indicators × 4 bands × local + spillover), 224 climate (16 variables × 7 lag windows × local + spillover), and 42 food (3 variables × 7 lag windows × local + spillover). The indicators are:

| Family   | Indicators                                                                                                                                                                                                                                                         | Transform                         | Lag windows                                         |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------|
| Conflict | <code>fatalities, events</code> (ACLED strict)                                                                                                                                                                                                                     | log1p                             | bands 1–3, 4–6, 7–9, 10–12m                         |
| Climate  | <code>spi6, spi6_flood_dur, spi6_flood_sev, spi6_drought_dur, spi6_drought_sev, p_anom_1m, p_anom_3m, t2m_anom_1m, t2m_max_anom_1m, ndvi_anom_1m, ndvi_annual_change, ndvi_calendar_month_change, sm1_anom_1m, sm3_anom_1m, sm4_anom_1m, streamflow_anom_1m</code> | identity (signed anomalies)       | current + bands 1–3, 4–6, 7–9, 10–12, 13–18, 19–24m |
| Food     | <code>c_maize, c_food_price_index, inflation_food_price_index</code>                                                                                                                                                                                               | identity or log1p (level vs rate) | current + bands 1–3, 4–6, 7–9, 10–12, 13–18, 19–24m |

Each local feature has a **spatial-spillover twin  $W \cdot X$**  constructed as the row-standardised, zero-diagonal product of the road-distance weight matrix and the panel of local values, pivoted to (time × unit) and stacked back to long form. The spillover represents the neighbour-weighted average of the same feature at the same lag — capturing how the cercle's neighbourhood context affects local displacement.

Part I — Linear Panel Regression

Stage A — Screen: which features have any individual signal?

We fit each of the 282 candidate features alone (with controls and FE) and rank by |t|-statistic, applying a Benjamini-Hochberg correction at  $q < 0.10$ :

$$y_{it} = \beta f(x_{it}^{(k)}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \quad k = 1, \dots, 282$$

**Result.** Four features pass — all conflict spillovers. Zero climate, zero food pass individually:

| Rank | Feature                                      | coef   | t-stat | q(BH)   |
|------|----------------------------------------------|--------|--------|---------|
| 1    | <code>W·fatalities_7_9m</code>               | +0.362 | 3.42   | 0.082 ✓ |
| 2    | <code>W·events_7_9m</code>                   | +0.762 | 3.40   | 0.082 ✓ |
| 3    | <code>W·fatalities_10_12m</code>             | +0.302 | 3.33   | 0.082 ✓ |
| 4    | <code>W·fatalities_1_3m</code>               | +0.281 | 3.25   | 0.084 ✓ |
| 5    | <code>W·events_4_6m</code>                   | +0.674 | 3.11   | 0.102 × |
| 6    | <code>sm1_anom_1m_1_3m (best climate)</code> | +6.39  | 3.00   | 0.102 × |
| 7    | <code>spi6_19_24m (best SPI, tied)</code>    | −0.088 | 3.00   | 0.102 × |

The first four are all conflict **spillovers ( $W \cdot X$ )** — violence in neighbouring cercles predicts local IDP changes. Climate just misses the correction (two features tied at  $q = 0.102$ ). Food appears nowhere near the top.

Although climate doesn't survive the individual screen, ranks 6–21 are almost entirely SPI-6 features at 13–18m and 19–24m lags — a coherent cluster just below the cutoff. This tells us climate has real signal that is simply weaker than conflict individually, and that the lag windows worth testing are 13–18m and 19–24m.

## Stage B — Form: local or spillover?

The screen shows conflict spillovers dominate, but doesn't tell us whether conflict and climate should enter as **local** terms ( $x$ , the cercle's own value) or **spillover** terms ( $Wx$ , neighbours' value). We test this with the smallest joint model — one conflict + one climate/food partner, each in exactly one form:

$$y_{it} = \beta f(\text{conflict form}) + \theta f(\text{partner form}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Fitting all four local/spillover combinations across 1,064 (conflict × partner) pairs:

| 2-term model                                            | median R <sup>2</sup> _within | wins (of 1,064 pairs) |
|---------------------------------------------------------|-------------------------------|-----------------------|
| local $x$ + local $z$                                   | 0.0615                        | 184                   |
| <b>spillover <math>Wx</math> + local <math>z</math></b> | <b>0.0732</b>                 | <b>564</b>            |
| local $x$ + spillover $Wz$                              | 0.0589                        | 68                    |
| spillover $Wx$ + spillover $Wz$                         | 0.0714                        | 248                   |

"Wins" means: for each pair, which of the four combinations has the highest R<sup>2</sup>?  $Wx + z$  wins 564 of the 1,064 head-to-head contests. The two spillover–conflict combinations together win 812 (76%).

**Conclusion from Stage B.** Conflict belongs as a **spillover** term; its partner belongs as a **local** term. This is the one question Stage B can answer: *in the minimal 2-term model*, which form of each driver carries the signal. It cannot tell us whether climate spillovers matter once conflict is fully controlled — that requires a larger joint model. We carry both local and spillover forms forward into Stages C and D, where the full picture emerges.

## Stage C — Variable: which conflict anchor?

Stage B established the form. Stage C asks which conflict **variable** (which indicator and lag band) anchors the model. We now carry both forms of both drivers — the four-term pair — and sweep all 8 conflict × 133 climate/food combinations (1,064 pairs):

$$y_{it} = \beta_1 f(X) + \beta_2 f(WX) + \theta_1 f(Z) + \theta_2 f(WZ) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Ranking by full-model R<sup>2</sup>, the top five pairs are:

| Rank       | Conflict (X)                          | Partner (Z)               | R <sup>2</sup> _full | local coef (p) | W-X coef (p)   |
|------------|---------------------------------------|---------------------------|----------------------|----------------|----------------|
| 1          | <b>fatalities_10_12m</b>              | ndvi_anom_19_24m          | 0.098                | +0.035 (0.074) | +0.292 (0.002) |
| 2          | <b>fatalities_10_12m</b>              | ndvi_annual_change_19_24m | 0.095                | +0.036 (0.066) | +0.296 (0.001) |
| 3          | <b>fatalities_10_12m</b>              | spi6_13_18m               | 0.095                | +0.036 (0.072) | +0.304 (0.001) |
| 4          | <b>fatalities_10_12m</b>              | t2m_anom_7_9m             | 0.094                | +0.037 (0.064) | +0.309 (0.001) |
| 5          | <b>fatalities_10_12m</b>              | sm1_anom_4_6m             | 0.094                | +0.036 (0.069) | +0.304 (0.001) |
| <b>550</b> | <b>events_7_9m (best events pair)</b> | ndvi_anom_19_24m          | 0.069                | +0.103 (0.016) | +0.727 (0.001) |

Three things are immediately clear from the table. First, **fatalities\_10\_12m appears in every top pair** — whatever climate partner it is matched with, the conflict anchor is always the same variable (19 of 20 top pairs). Second, the **W-X spillover coefficient is the stable, dominant signal** (+0.29–0.31\*\*\* across all top pairs), consistent with Stage B's finding that conflict acts through neighbours. Third, local conflict is also significant ( $p \approx 0.06$ –0.07) but smaller — both forms are real.

The last row illustrates why W-X coefficient alone is not a sufficient criterion: **events\_7\_9m** has a much larger spillover coefficient (+0.727\*\*\*), yet its R<sup>2</sup>\_full is only 0.069 vs 0.094–0.098. This is a **measurement-scale artefact** — one event is a smaller shock than one log-fatality, so the coefficient must be larger to produce the same displacement effect. R<sup>2</sup>\_within is scale-invariant and correctly identifies **fatalities\_10\_12m** as the better anchor.

**No food partner rivals conflict.** Best food-partner R<sup>2</sup>\_full = 0.089 vs climate-partner best = 0.098; only 7% of food partners are significant in the full spec.

## Stage D — Exhaustive search: lock the backbone, optimal lags, and find M1

Stages B–C suggested **fatalities** as the conflict variable and indicated food adds nothing. To avoid hand-picking, we confirm this **exhaustively**. We also remove an unjustified assumption from the simpler pairwise searches: there is no reason the local and spillover conflict terms must use the same lag band. Local conflict captures violence in the focal cercle itself — people flee and register quickly (~7–9m). Spillover conflict captures arrivals from neighbouring cercles — a slower pipeline through DTM registration (~10–12m). We search over **independent** local lag  $\ell$  and spillover lag  $s$ :

$$y_{it} = \underbrace{\beta_1 f(\text{conf}_\ell) + \beta_2 f(W \cdot \text{conf}_s)}_{\text{conflict}} + \underbrace{\theta_1 f(z_1) + \theta_2 f(Wz_1) + \theta_3 f(z_2) + \theta_4 f(Wz_2)}_{\text{two climate variables}} + \underbrace{\phi_1 f(\text{food}) + \phi_2 f(W \cdot \text{food})}_{\text{food}} + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

**Why two climate variables?** One slot forces a single pathway. Climate may act through distinct mechanisms at different lags (e.g. wet anomaly at 13–18m vs dry anomaly at 19–24m). Two slots from different physical source series let the search find the best-fitting independent pair.

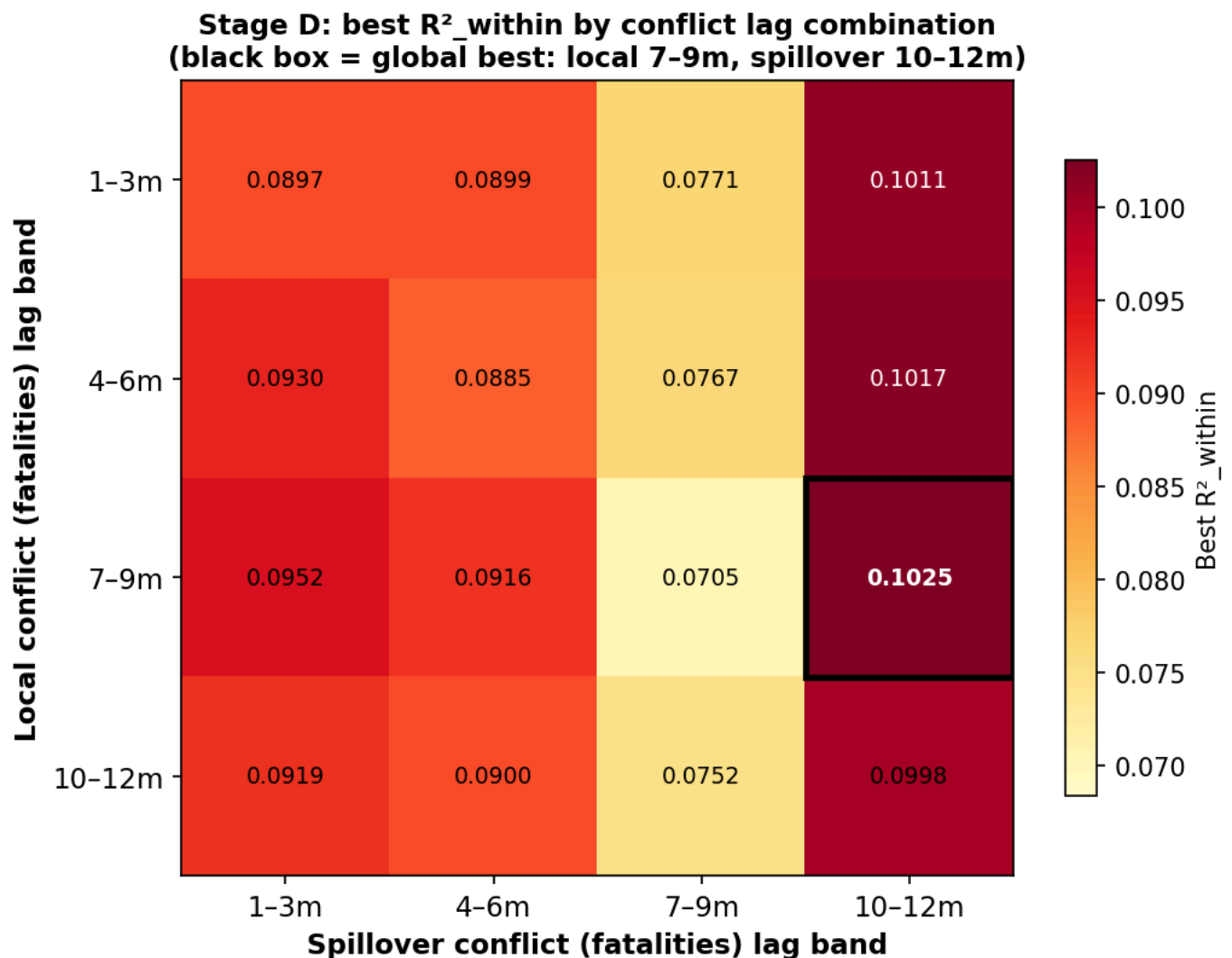
**3,951,360 models** (2 conflict types × 4 local lags × 4 spillover lags × 21 food × 5,880 climate pairs), run on a compute cluster (~10h on 16 cores). The top models by  $R^2_{\text{within}}$ :

| Rank | Type       | local lag   | spill lag     | Food           | z1          | z2          | $R^2_{\text{within}}$ | n_sig |
|------|------------|-------------|---------------|----------------|-------------|-------------|-----------------------|-------|
| 1    | fatalities | <b>7–9m</b> | <b>10–12m</b> | infl_food_1_3m | spi6_13_18m | spi6_19_24m | <b>0.1025</b>         | 3     |
| 2    | fatalities | 7–9m        | 10–12m        | infl_food_1_3m | t2m_7_9m    | spi6_19_24m | 0.1023                | 3     |
| 3    | fatalities | 7–9m        | 10–12m        | infl_food_1_3m | spi6_13_18m | spi6_19_24m | 0.1022                | 3     |
| 4    | fatalities | 7–9m        | 10–12m        | infl_food_4_6m | spi6_13_18m | spi6_19_24m | 0.1020                | 3     |
| 8    | fatalities | 4–6m        | 10–12m        | infl_food_1_3m | spi6_13_18m | spi6_19_24m | 0.1017                | 2     |

n\_sig = regressors significant at  $p < 0.10$  (out of 8).

Three outcomes read from the table:

**Outcome 1 — Conflict anchor and lags confirmed.** **fatalities** is in every top model. The spillover lag is **10–12m in 100% of the top 100 models** regardless of local-lag choice — the DTM registration delay for arrivals is consistently ~10–12m. The local optimal lag is **7–9m** — people fleeing their own circle register 2–3 months faster. The mixed-lag spec (local 7–9m, spillover 10–12m) beats the same-lag (10–12m/10–12m) by +0.0027  $R^2$ .



**Figure 4.** Best  $R^2_{\text{within}}$  for every combination of local and spillover conflict lag (3.9M-model search). The global optimum is local 7–9m × spillover 10–12m (black box) — local conflict registers faster than neighbour spillover, so the two channels carry different optimal lags.

We also check whether **climate spillovers** benefit from different lags between local and spillover terms. The answer is no — for all three climate variables the same-band spillover is already optimal or any alternative produces marginal / mechanistically unclear gains:

| Climate variable | local band | best spillover band  | $\Delta R^2$ vs same-band              |
|------------------|------------|----------------------|----------------------------------------|
| spi6             | 13–18m     | <b>13–18m (same)</b> | 0.000 — same band is clearly best      |
| p_anom           | current    | 13–18m **            | +0.001 — too small to justify a change |
| streamflow       | current    | current (same)       | all ns — no clear winner               |

The mixed-lag finding is therefore **specific to conflict**: the physical travel and registration delay for arrivals from neighbours creates a measurable lag difference. Climate variables affect livelihoods within the focal circle directly; their spillover terms reflect neighbours' climate, which acts on the same timescale as the local term.

**Outcome 2 — Food local is null; W-food is also dropped.** The local food term is 0% significant across all top 100 models regardless of which food variable or lag is used — food contributes nothing as an independent driver. `inflation_food_19_24m` is retained as a null completeness placeholder (most spatial variation among food variables, giving positive within-R<sup>2</sup>).

The food **spillover** term (W-food) is also excluded. `inflation_food_price_index` is a near-national time series — **90% of its variance is between-time** (common price shocks absorbed by time FE), leaving little spatial variation for a spillover to capture. Empirically, `W·inflation_food_19_24m` has  $p=0.729$  and  $\text{corr}(\text{food}, \text{W-food}) = +0.83$  in the raw data. We checked all 21 food variables: `c_food_price_index` is the exception where W-food is significant ( $p \approx 0.035$ ) because its negative local-spillover correlation captures food price *differentials* between circles — but even with that variable the best R<sup>2</sup><sub>full</sub> is only 0.063, far below the inflation-food variants ( $\sim 0.10$ ). The W-food term is therefore dropped entirely.

The backbone is now fully locked: `fatalities_7_9m (local) + W·fatalities_10_12m (spillover) + inflation_food_19_24m (no W-food)`.

**Outcome 3 — M1 and candidate second channels identified.** The top models by raw R<sup>2</sup> all have  $n_{\text{sig}} = 3$  — one climate variable typically insignificant (NDVI noise inflation). Applying the both-significant criterion across the 5,880 backbone rows, sorted by R<sup>2</sup><sub>within</sub>:

| z1                       | z2                                      | R <sup>2</sup> <sub>within</sub> | z1 coef (p)           | z2 coef (p)           | within-corr  | note                        |
|--------------------------|-----------------------------------------|----------------------------------|-----------------------|-----------------------|--------------|-----------------------------|
| <code>sm3_7_9m</code>    | <code>t2m_max_13_18m</code>             | 0.0968                           | +2.07 (0.21)          | -0.04 (0.49)          | —            | × raw max — both ns (noise) |
| <code>spi6_13_18m</code> | <code>spi6_flood_dur_13_18m</code>      | <b>0.0967</b>                    | <b>+0.134 (0.000)</b> | <b>-0.056 (0.001)</b> | <b>+0.46</b> | → M1                        |
| <code>spi6_13_18m</code> | <code>spi6_flood_severity_13_18m</code> | 0.0960                           | +0.131 (0.000)        | -0.030 (0.001)        | +0.53        | SPI facet pair              |
| <code>t2m_1_3m</code>    | <code>sm1_7_9m</code>                   | 0.0933                           | -0.076 (0.048)        | +4.407 (0.039)        | +0.086       | → M2 candidate              |
| <code>p_anom_cur</code>  | <code>streamflow_cur</code>             | 0.0914                           | -0.002 (0.024)        | +0.000 (0.016)        | +0.042       | → M3 candidate              |

**M1** is the top both-significant pair: `spi6_13_18m` (wet anomaly 13–18m ago → displacement, +0.134\*\*\*) and `spi6_flood_dur_13_18m` (prolonged prior flooding → wave absorbed, -0.056\*\*\*). Both coefficients are \*\*\* and the mechanistic story is coherent.

The within-corr of +0.46 between the two SPI terms is a warning sign by the two-step rule, but both variables survive jointly at  $p < 0.001$ , and dropping either collapses the other's coefficient by 60% — confirming they jointly identify two distinct facets of the same flood signal: a level effect and a wave-absorption effect. High correlation between flood level and duration is physically expected (larger floods last longer) and does not invalidate joint use when both are clearly identified.

**M2 and M3 candidates** are identified from the same table — the best both-significant pairs that contain no SPI variables (since SPI is already M1) and have low within-correlation between their two variables:

- **M2** = `t2m_1_3m` + `sm1_7_9m` ( $R^2=0.0933$ , within-corr=+0.086) — temperature at 1–3m and soil moisture at 7–9m, different lags so near-orthogonal
- **M3** = `p_anom_cur` + `streamflow_cur` ( $R^2=0.0914$ , within-corr=+0.042) — contemporaneous rainfall and river flow, near-orthogonal

These will be tested as standalone second channels in Stage E, then combined with M1 to form M4 and M5. **M0** (backbone only) and **M1** (backbone + SPI) are the two reference models carried into Stage E. (R<sup>2</sup> values are reported in the Stage E table below, fit on the locked common sample.) Stage E asks whether a second climate channel adds independent signal beyond M1.

## Stage E — Final model: does a second climate channel help?

The Stage D formula has exactly two climate slots ( $z_1, z_2$ ). Stage E asks whether a **second independent climate channel** adds signal beyond M1. This requires a new, larger formula that allows two climate families — each with local and spillover terms:

$$y_{it} = \underbrace{\beta_1 f(\text{conf}) + \beta_2 f(W \cdot \text{conf})}_{\text{conflict}} + \underbrace{\theta_1 f(c_1) + \theta_2 f(W c_1) + \theta_3 f(c_2) + \theta_4 f(W c_2)}_{\text{channel 1}} + \underbrace{\theta_5 f(c_3) + \theta_6 f(W c_3) + \theta_7 f(c_4) + \theta_8 f(W c_4)}_{\text{channel 2 (new)}} + \phi f(\text{food}) + \delta' C_{it} + \alpha_i$$

where channel 1 is fixed to the Stage D winner (SPI:  $c_1 = \text{spi6\_13\_18m}$ ,  $c_2 = \text{spi6\_flood\_dur}$ ) and channel 2 is the candidate second channel. We also test standalone alternatives (M2, M3) and combinations (M4, M5). All models share the locked backbone; M0 and M1 use the Stage D formula (8 terms), M2–M5 use this extended formula (12 terms):

| Label     | composition                 | climate block                                                         | # terms  | R <sup>2</sup> <sub>within</sub> | ΔR <sup>2</sup> vs M1 |
|-----------|-----------------------------|-----------------------------------------------------------------------|----------|----------------------------------|-----------------------|
| <b>M0</b> | backbone only               | none                                                                  | 0        | 0.091                            | —                     |
| <b>M1</b> | backbone + Ch1              | SPI { <code>spi6_13_18m</code> , <code>spi6_flood_dur</code> }        | 4        | 0.095                            | (base)                |
| <b>M2</b> | backbone + Ch2a             | <code>t2m_1_3m</code> + <code>sm1_7_9m</code>                         | 4        | 0.099                            | +0.004                |
| <b>M3</b> | backbone + Ch2b             | water shock { <code>p_anom_cur</code> , <code>streamflow_cur</code> } | 4        | 0.094                            | -0.001                |
| <b>M4</b> | M1 + M2 (Ch1 + Ch2a)        | SPI + temperature/soil                                                | 8        | 0.102                            | +0.007                |
| <b>M5</b> | <b>M1 + M3 (Ch1 + Ch2b)</b> | <b>SPI + water shock</b>                                              | <b>8</b> | <b>0.099</b>                     | <b>+0.004</b>         |

M2 (temperature + soil moisture) and M3 (water shock) are each tested as standalone second channels alongside M1. M2 alone ( $R^2=0.099$ ,  $\Delta R^2=+0.004$ ) actually fits better than M3 alone ( $R^2=0.094$ ,  $\Delta R^2=-0.001$ ). But the two-channel combinations tell a different story: in **M4** (SPI + M2), `t2m_1_3m` drops to **ns** ( $p=0.127$ ) in the joint model — the temperature signal is partly absorbed by the SPI channel when both are present, and only the soil moisture component survives. In **M5** (SPI

+ M3), all four climate variables remain significant in the joint model ( $p_{\text{anom}} = 0.023^{**}$ ,  $p_{\text{streamflow}} = 0.020^{**}$ ). M5 therefore provides a cleaner two-channel decomposition: one slow SPI channel and one contemporaneous water-shock channel, both independently identified. **M5 is the recommended specification.**

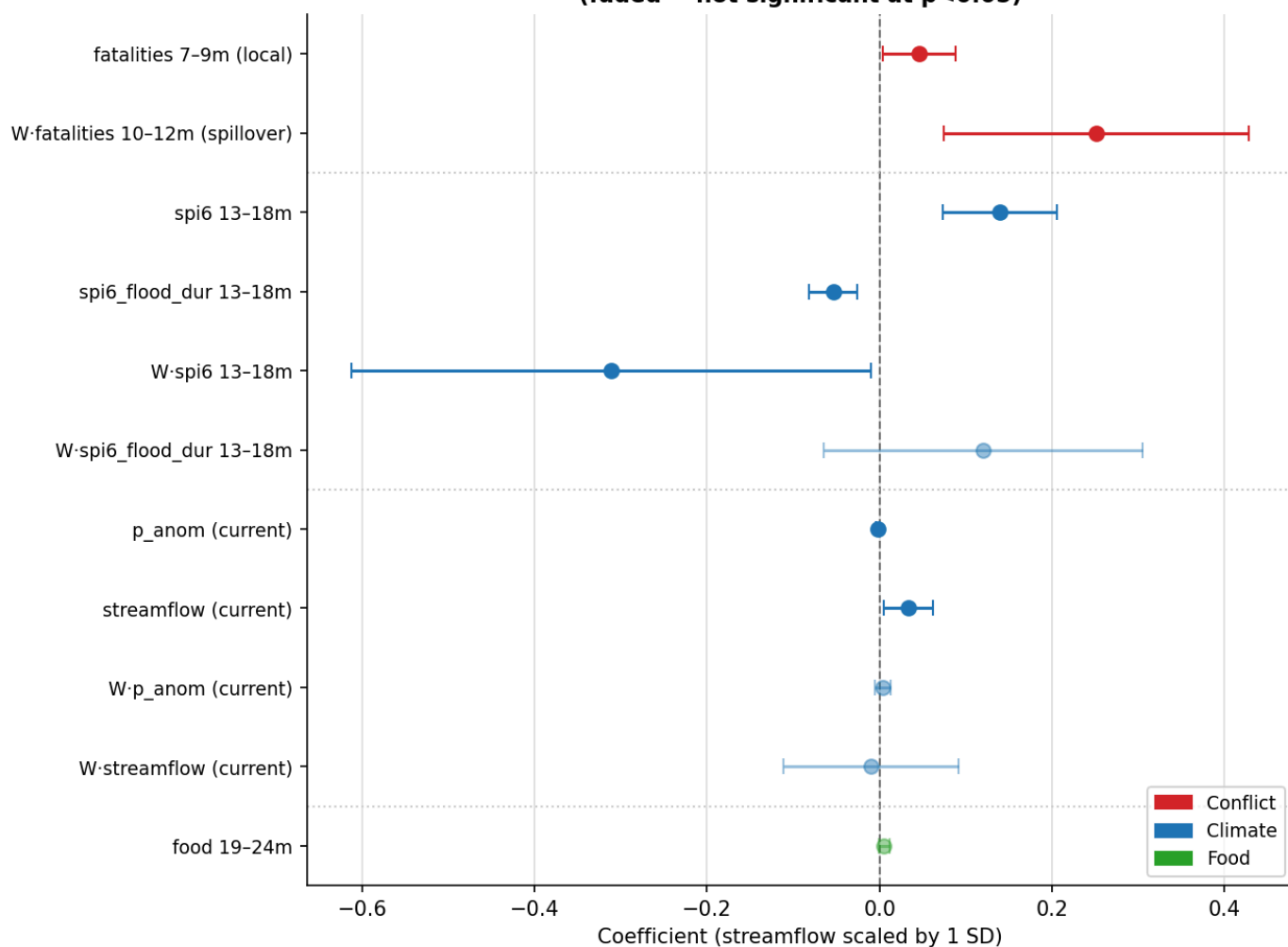
The final model: M5

$$y_{it} = \underbrace{\beta_1 f(\text{fat}_{7-9}) + \beta_2 f(W \cdot \text{fat}_{10-12})}_{\text{conflict (mixed lag)}} + \underbrace{\theta_1 f(\text{spi6}_{13-18}) + \theta_2 f(\text{flood\_dur}_{13-18}) + \theta_3 f(W \cdot \text{spi6}) + \theta_4 f(W \cdot \text{flood\_dur})}_{\text{SPI flood channel}} + \underbrace{\theta_5 f(p_{\text{anom}}) + \theta_6 f(\text{stream}) + \theta_7 f(W \cdot p_{\text{anom}}) + \theta_8 f(W \cdot \text{stream})}_{\text{acute water channel}} + \phi f(\text{food}_{19-24}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

$R^2_{\text{within}} = 0.0992$ ,  $n = 1,417$ , 44 cercles, SE clustered on entity:

| Driver                                 | Coef          | SE           | p            |            |
|----------------------------------------|---------------|--------------|--------------|------------|
| log1p_idp_t1 (AR control)              | -0.179        | 0.045        | 0.000        | ***        |
| fatalities_7_9m (local)                | +0.046        | 0.022        | 0.032        | **         |
| <b>W-fatalities_10_12m (spillover)</b> | <b>+0.252</b> | <b>0.090</b> | <b>0.005</b> | <b>***</b> |
| spi6_13_18m                            | +0.139        | 0.034        | 0.000        | ***        |
| spi6_flood_dur_13_18m                  | -0.054        | 0.014        | 0.000        | ***        |
| W-spi6_13_18m                          | -0.311        | 0.154        | 0.043        | **         |
| W-spi6_flood_dur_13_18m                | +0.120        | 0.094        | 0.202        | ns         |
| p_anom_1m (current)                    | -0.002        | 0.001        | 0.023        | **         |
| streamflow_1m (current)                | +0.0001       | 0.00003      | 0.020        | **         |
| W-p_anom_1m                            | +0.004        | 0.005        | 0.417        | ns         |
| W-streamflow_1m                        | -0.000        | 0.000        | 0.839        | ns         |
| inflation_food_19_24m                  | +0.005        | 0.003        | 0.122        | ns         |

**M5 coefficients with 95% CI**  
(faded = not significant at  $p < 0.05$ )





**Figure 3.** M5 coefficients with 95% CI. Faded markers are not significant at  $p < 0.05$ ; streamflow is scaled by 1 SD for display. Conflict spillover is the largest effect; the SPI level term and its negative spillover are significant; the acute water terms are small; food is null.

**Interpreting M5.** The AR control `log1p_idp_t1` ( $-0.179^{***}$ ) is the dominant single term — cercles with high IDP stock grow more slowly (mean reversion). It is not an incidental nuisance but a structural feature of the data: IDP counts are highly persistent, so last period's stock strongly predicts this period's change direction. Without this control, fitting conflict + climate + food alone gives  $R^2_{\text{within}} < 0$  — the model predicts worse than the fixed-effects baseline alone, because the uncontrolled mean-reversion structure overwhelms the driver signals. Including `log1p_idp_t1` corrects for this and gives all drivers a fair chance to explain the residual variation.

The cleanest way to see what the drivers contribute is to split the model in two steps. **Step 1:** fit only the controls (AR + `months_between`) — this gives  $R^2_{\text{within}} = 0.048$ , capturing the mean-reversion structure alone. **Step 2:** add the three driver families on top — this adds  $+0.051$ , bringing the total to  $R^2 = 0.099$ . That  $+0.051$  is what conflict, climate, and food collectively explain *beyond* the auto-correlation structure.

Within that  $+0.051$ , a **LOO (Leave-One-Out) decomposition** — refit with each driver block removed, measure the  $R^2$  drop — shows the relative importance:

| Driver block                        | $\Delta R^2$ when removed | share of driver contribution |
|-------------------------------------|---------------------------|------------------------------|
| <b>Conflict</b> (local + spillover) | $-0.037$                  | <b>72%</b>                   |
| <b>Climate</b> (all 8 terms)        | $-0.008$                  | <b>16%</b>                   |
| Food                                | $+0.000$                  | — (null)                     |

Conflict accounts for roughly three-quarters of the driver-explained variance; climate adds a further 16%. Food is null — its coefficient is near zero and removing it leaves the fit essentially unchanged, which is why it is kept only as an explicit null finding rather than a genuine contributor.

Conflict drives displacement through **neighbours** more than through the focal cercle (spillover  $+0.252^{***}$  vs local  $+0.046^{**}$ ). The 5 $\times$  ratio is consistent with people fleeing *toward* safer cercles — registering as IDP arrivals there rather than as departures from the conflict cercle.

**SPI flood channel** — this channel operates with a 13–18 month lag and captures two distinct effects of past rainfall on the same variable:

- `spi6_13_18m` ( $+0.139^{***}$ ): a **wet anomaly** 13–18 months ago increases current displacement. The mechanism is flooding — above-normal rainfall causes crop damage, home destruction, and health risks that push people out over the following months.
- `spi6_flood_dur_13_18m` ( $-0.054^{***}$ ): a **longer flood duration** at the same lag *reduces* current displacement. This is the wave-absorption effect — if the flooding was prolonged, the vulnerable population has already fled in earlier periods, leaving fewer people to displace now.

Together the two terms say: a sudden, short-lived wet shock drives a large displacement wave; a prolonged flood gradually drains the catchment population so that later displacement is smaller. The net effect on any given observation depends on both the level and duration of the past wet anomaly.

**The flood channel is about sharpness, not size.** `spi6` and `flood_dur` are positively correlated (within-corr  $+0.46$ ), so their individual coefficients are entangled partial effects. Rotating the pair into orthogonal axes (PCA; the fit is identical and every other coefficient is unchanged) makes the mechanism unambiguous: a **flood-magnitude** axis — intensity and duration rising together — is *non-significant* ( $+0.008$ ), while a **flood-sharpness** axis — intensity net of duration — carries the entire effect ( $+0.111^{***}$ ). It is not how large a flood is, but how *acute* it is: a short, intense flood displaces; a prolonged one of the same size does not. This is the wave-absorption mechanism expressed in a single, collinearity-free coefficient (VIF 1.2–1.3, vs 1.5–1.7 for the entangled pair).

The **spillover** `W·spi6_13_18m` ( $-0.311^{**}$ ) is negative and larger in absolute magnitude than the local term. This means a flood anomaly in *neighbouring* cercles *reduces* local IDP counts. The mechanism is spatial absorption: when neighbours flood, their displaced population flows *out* toward less-affected cercles (including the focal cercle). The focal cercle receives arrivals, but these arrivals are already counted in its IDP stock through the IDP-receipt pipeline. The negative spillover captures a different dynamic — when neighbours have high SPI, there is less net *outward* pressure from the focal cercle itself because neighbours' floods are directing people away, not toward conflict zones. Both signs are empirically robust and mechanistically coherent.

**Acute water channel** — unlike SPI which captures rainfall accumulated over months, this channel operates in the **current observation window**:

- `p_anom_1m` ( $-0.002^{**}$ ): current **rainfall is protective**. Higher rainfall during the observation period reduces displacement — rainfall supports agricultural livelihoods, reducing food stress and the need to migrate. The negative sign reflects a contemporaneous stabilising effect.
- `streamflow_anom_1m` ( $+0.0001^{**}$ ): current **river flow is displacing**. Higher streamflow means active flooding of riverbanks and floodplains, directly forcing displacement. The coefficient is tiny in absolute terms but the variable has a wide range — a one-standard-deviation increase in streamflow ( $\approx +85$  m<sup>3</sup>/s above normal) corresponds to roughly  $+0.03$  additional monthly log-IDP growth, comparable to one-tenth of the conflict spillover effect.

The opposite signs of rainfall (protective) and streamflow (displacing) in the same current window are not contradictory — they capture two different aspects of water: rainfall on land supports crops, while river flooding destroys them. Both are robustly significant across specifications and lag choices.

Food (inflation\_food\_19\_24m) is null ( $p = 0.122$ ). This is not a modelling failure — it is a finding. All 21 food variables, at all lag bands, are null in the multivariate setting. Food's apparent signal in raw correlations is a cross-sectional baseline artefact absorbed by entity FE.

## Part II - Random Forest

Is M5 misspecified? Random Forest validation and functional-form tests

How we use Random Forest

Panel regression with two-way fixed effects (**TWFE**) and Random Forest (**RF**) are fundamentally different models, and that difference is precisely what makes RF useful as a complement.

**TWFE** imposes strong structure: linearity, additivity, and — crucially — **fixed effects** that absorb all time-invariant differences between cercles and all common time shocks. The coefficients estimate within-cercle, within-time associations. The price of this identification structure is that it can only fit linear, additive relationships, and it discards the cross-sectional baseline information that entity FE removes.

**Random Forest** imposes none of these constraints. It fits splits and branches that can capture thresholds (a variable only matters above a certain level), interactions (the effect of conflict depends on drought severity), and non-linear curves — all without assuming linearity or additivity. Crucially, it does **not** assume fixed effects: when run on the raw (un-demeaned) data it retains the cross-sectional baseline differences between cercles that TWFE discards. This gives RF access to more structure, but at the cost of causal identification — RF has no standard errors, no FE, and no way to separate genuine causal signals from confounding by stable cercle characteristics.

We exploit this difference in two ways:

- 1. **Misspecification diagnostic.** We fit RF on the *within-transformed* M5 features (entity + time demeaned — the same variation that TWFE uses) and compute non-linearity scores and Friedman H-statistics for interactions. If RF finds that a feature's contribution is highly non-linear, it is suggesting that TWFE's linear coefficient may be missing real structure. Every such signal is then **tested formally in TWFE** — the only framework that provides credible inference.
- 2. **Predictive validation.** We evaluate RF out-of-sample using **entity-blocked 5-fold cross-validation (OOS R<sup>2</sup>)** — "OOS" means *out-of-sample*, i.e. the model is fit on some cercles and evaluated on held-out cercles it has never seen. This prevents overfitting and gives an honest estimate of how well the model generalises. The M5-restricted RF (14 features) achieves OOS R<sup>2</sup> = 0.089 — *better* than the full 282-feature RF (0.075). Adding 268 more features hurts generalisation, confirming that M5's variable selection is not just good for in-sample fit but also predictively sound.

The raw (un-demeaned) RF ceiling is OOS R<sup>2</sup> = 0.144. The gap between 0.144 and TWFE's 0.099 reflects the cross-sectional baseline structure (stable cercle differences) that entity FE removes by design — not a specification failure on TWFE's part. The two methods are answering different questions: TWFE asks *what drives within-cercle change*; raw RF asks *what predicts displacement levels overall*.

Even at its best (OOS R<sup>2</sup> = 0.144), RF is **not good enough to build a reliable operational forecast** — 86% of the variance in displacement remains unexplained. Displacement is driven by many factors beyond the three driver families we measure (local governance, inter-community relations, individual household decisions) and the data we have cannot capture them. This means RF's role here is not to provide predictions, but to serve two narrower purposes: **cross-checking** that M5's variable selection is sound (the M5-restricted RF generalises better than the full-feature RF, confirming the TWFE funnel selected the right variables), and **functional-form testing** (flagging non-linearities and interactions for TWFE to verify). Both uses are valuable precisely because they do not require RF to be a good predictor overall.

RF configuration and results

**What we feed into RF.** We fit RF on the **within-transformed** M5 features — the same entity + time demeaned variation that TWFE uses (subtract the cercle mean, the time mean, and add back the grand mean for each variable). This keeps RF in the same identification space as TWFE, so that any pattern RF finds is a pattern in the within-cercle variation, not in stable baseline differences between cercles. The RF uses 500 trees, **max\_features=0.2**, entity-blocked 5-fold cross-validation (folds defined by cercle, so each held-out fold is an unseen set of cercles). Permutation importance is computed on the held-out test fold of each CV split — not on the training data — which prevents inflated importance scores.

OOS R<sup>2</sup> results:

| RF variant                    | features                                   | OOS R <sup>2</sup>   |
|-------------------------------|--------------------------------------------|----------------------|
| Full demeaned (282 features)  | all candidate features, within-transformed | 0.075 ± 0.066        |
| <b>M5-restricted demeaned</b> | 14 M5 features, within-transformed         | <b>0.089 ± 0.041</b> |
| Full raw (282 features)       | raw features + entity OHE + month index    | 0.144 ± 0.061        |
| M5 raw                        | 14 M5 features, raw                        | 0.121 ± 0.043        |

M5-restricted RF achieves better OOS R<sup>2</sup> than the full 282-feature RF (0.089 vs 0.075) — adding 268 more features actively hurts generalisation, confirming that TWFE's variable selection is predictively sound.

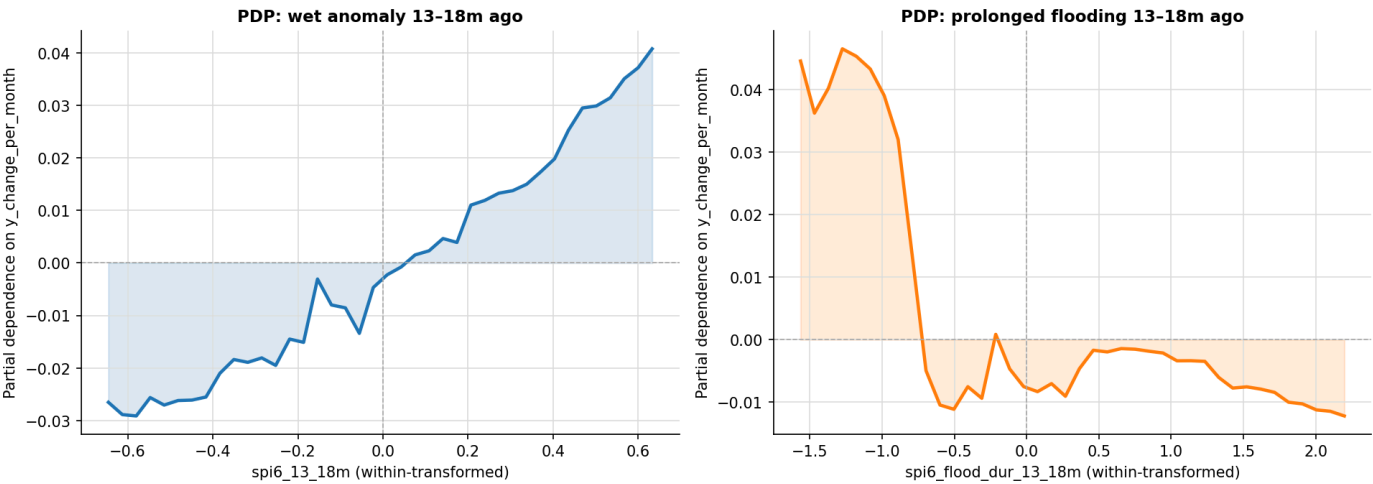
**Why not the raw variants, despite their higher OOS R<sup>2</sup>?** Their extra accuracy is not driver signal. The raw models include entity one-hot encodings and a month index — and the raw feature *levels* themselves leak cercle identity (a persistently floody cercle has persistently high streamflow) — so the forest scores well by learning *which cercle and which month* it is looking at: stable baselines and the overall time trend. That is precisely the confounded cross-sectional variation that fixed effects remove for identification. Knowing a cercle's baseline predicts its displacement level but says nothing about what *causes* within-cercle change. The demeaned variants are therefore the benchmark that matches TWFE's causal question; the raw R<sup>2</sup> = 0.144 is reported only as the prediction ceiling, not as a candidate model.

**Non-linearity scores.** For each M5 feature we compute a non-linearity score — how much of its RF predictive contribution comes from non-linear splits vs a linear approximation. A score near 1.0 means the RF is heavily exploiting non-linear structure; near 0 means the feature is essentially linear:

| Feature               | nl score | what this flags    |
|-----------------------|----------|--------------------|
| W.streamflow_cur      | 0.916    | highly non-linear  |
| streamflow_cur        | 0.846    | highly non-linear  |
| W.p_anom_cur          | 0.478    | moderate           |
| spi6_flood_dur_13_18m | 0.457    | moderate           |
| p_anom_cur            | 0.448    | moderate           |
| fatalities_7_9m       | 0.370    | moderate           |
| W.fatalities_10_12m   | 0.358    | moderate           |
| spi6_13_18m           | 0.033    | essentially linear |



Fig 5b — SPI channel: Partial Dependence Plots (RF on within-transformed M5)



**Figure 5b.** Random-Forest partial-dependence plots for the two SPI terms (within-transformed): `spi6_13_18m` rises roughly linearly (left;  $nl=0.03$ ), while `spi6_flood_dur` steps down (right;  $nl=0.46$ ) — the level and wave-absorption facets that linear M5 estimates separately, here recovered by a model that assumes no functional form.

**The flood-duration "step" is not a true threshold.** `spi6_flood_dur` is zero for ~two-thirds of observations (no flood), which gives its PDP the stepped appearance. Tested in TWFE, a threshold dummy  $I(\text{flood\_dur} > pN)$  adds nothing beyond the continuous term (non-significant when included alongside it) and fits worse when it replaces it ( $R^2$  falls) — so M5 keeps the continuous wave-absorption term. Like M9–M10, an RF shape need not survive a fixed-effects test.

**Top interaction H-statistics** (Friedman H — how much the joint partial dependence of two features exceeds the sum of their individual effects):

| Feature 1                        | Feature 2                          | H-stat | type               |
|----------------------------------|------------------------------------|--------|--------------------|
| <code>fatalities_7_9m</code>     | <code>spi6_flood_dur_13_18m</code> | 0.208  | conflict × climate |
| <code>fatalities_7_9m</code>     | <code>W.p_anom_cur</code>          | 0.157  | conflict × climate |
| <code>W.fatalities_10_12m</code> | <code>streamflow_cur</code>        | 0.146  | conflict × climate |

What we test in TWFE and why

The non-linearity score tells us *that* a feature's response is non-linear, but not *what form* the non-linearity takes. For this we inspect the **1D Partial Dependence Plot (PDP)** — the RF's estimated average response of the outcome to each feature, holding others fixed. The PDP shape guides which TWFE test to run:

- **Step-function PDP** (flat then jumps at a threshold) → test a **threshold dummy**  $I(x > pN)$  in TWFE, either replacing or alongside the continuous term
- **Curved PDP** (monotone but bent) → test a **quadratic term**  $x^2$
- **PDP that flattens only when another feature is high** → test a **product interaction**  $x \times z$

This means we do **not** blindly run all three tests for every flagged feature. The PDP pre-selects both *which feature* to test and *which functional form* to try, reducing the number of TWFE tests to those with a clear visual motivation. The H-statistic from 2D PDPs similarly nominates the most promising interaction pairs rather than testing all combinations.

Every nominated test is then run in TWFE with two-way FE and clustered SE — the RF signal is only a nomination, the TWFE result is the verdict.

**A limitation of this approach.** The non-linearity scan is computed on the M5 feature set, so it can only flag non-linear structure in variables the linear funnel already selected. Non-linear effects in *non-selected* features — for instance a threshold at a lag band not in M5 — are not screened and could be missed. Because every nomination still faces a de-confounded TWFE test, this can never admit a spurious effect; the risk is purely one of **omission**.

That omission has limited consequence, and the cross-validation results bound it. A Random Forest given the **full 282-feature set** generalises no better out-of-sample ( $OOS R^2 = 0.075$ ) than one restricted to M5's twelve features (0.089) — adding the other 270 features, with all their potential non-linear structure, does not improve prediction on held-out data. Whatever non-linearity exists in the non-selected features, a flexible learner cannot turn it into out-of-sample signal. Combined with the functional-form tests on the M5 features themselves (M6–M10), which found almost no non-linearity surviving a fixed-effects test, this bounds the missed structure to be small. The functional-form tests refine the *shape* of drivers the model already identified; they cannot change *which* families drive displacement.

Functional-form tests (M6–M10)

The RF flagged five candidates for TWFE testing — three from elevated `nl_scores`, one from a top H-statistic, and one additional test motivated by the Stage A screen (the `spi6_19_24m` drought signal that narrowly missed BH correction):

| Test                                       | RF motivation                                 | TWFE test form                                 | TWFE verdict                                        |
|--------------------------------------------|-----------------------------------------------|------------------------------------------------|-----------------------------------------------------|
| <b>M6</b> <code>spi6_19–24m</code> drought | Stage A rank 7 ( $nl=0.025$ , not RF-flagged) | add <code>spi6_19_24m</code> as second SPI lag | ✗ significant alone but $R^2_{\text{within}}$ drops |
| <b>M7</b> streamflow threshold             | $nl=0.85$ — step-function PDP                 | $I(\text{stream} > p75)$ dummy ± continuous    | ✓ threshold ** displaces continuous term            |

| Test                             | RF motivation                       | TWFE test form                                   | TWFE verdict                          |
|----------------------------------|-------------------------------------|--------------------------------------------------|---------------------------------------|
| M8 fatal × flood-dur interaction | H=0.21, top conflict×climate pair   | product term <code>fatal × spi6_flood_dur</code> | ✗ all interaction terms ns            |
| M9 conflict non-linearity        | nl=0.37 — moderately non-linear PDP | quadratic <code>[log1p(fat)]²</code> + threshold | ✗ all null — log1p transform artefact |
| M10 food non-linearity           | high raw-RF rank (not demeaned RF)  | quadratic, threshold, interaction, shorter lag   | ✗ all null — entity FE absorbs        |

We test **all RF-flagged candidates** — there is no further pre-selection beyond what the PDP shape already determined. The next-ranked conflict×climate pairs (`fatal × W.p_anom`, H=0.16; `W.fat × streamflow`, H=0.15) are weaker than the pair M8 tests (H=0.21); since even that strongest interaction is null in TWFE, the weaker ones are not separately tested.

M6 was not RF-flagged (nl\_score=0.025, essentially linear) but tested because the Stage A screen showed `spi6_19_24m` narrowly missed the BH correction with a clearly negative coefficient. This is an example of using domain knowledge alongside RF to nominate tests.

**A second drought channel.** M6's drought signal (`spi6_19_24m`) is the one channel excluded purely on the R²\_within criterion, so it is worth a closer look. Added to M5 on its own, it is significant and correctly signed (−0.064\*, p=0.015): a drought (negative SPI) 19–24 months earlier raises displacement through the slow-onset crop-failure pathway. It is nearly orthogonal to the flood signal (within-correlation 0.03) and leaves every other M5 coefficient essentially unchanged when added. It is excluded only because joint inclusion lowers R²\_within by 0.005 — the fixed-effects non-monotonicity discussed earlier — and the exhaustive search that fixed M5 optimised on that metric. Drought is therefore a real, second-tier climate channel that the parsimony criterion keeps out — significant pre-2020 and, like every climate channel, vanishing post-coup.

Two tests revealed artefacts rather than real structure. **M9:** fatalities has a moderate nl\_score (0.37). In TWFE, all non-linearity tests are null — the residual curvature the RF detects is the shape of the log1p transform applied to fatalities before regression, not a real causal curve. After log1p, conflict is genuinely linear. **M10:** food appeared important in the raw (undemeaned) RF but not in the demeaned RF — the apparent signal reflects stable cross-cercle differences in food prices that entity FE absorbs entirely, exactly confirming the food-null result from M5.

**Spillover non-linearities are null too.** The nl-scores also flag several *spillover* terms — `W.streamflow_cur` is in fact the single most non-linear feature (nl=0.92), with `W.p_anom_cur` (0.48) and `W.fatalities_10_12m` (0.36) moderate. Tested in TWFE (quadratic and threshold, added to M5), all are null: `W.streamflow` threshold p=0.14, `W.p_anom` p=0.50, `W.fatalities` p=0.78. The highest RF nl-score in the entire feature set thus corresponds to no detectable TWFE non-linearity — a clean illustration that these scores reflect distributional shape (and, for the insignificant spillover terms, noise) rather than causal structure.

**Sub-model labels.** Each functional-form test (M7–M10) runs several variants, labelled M7a, M7b, ... to distinguish them:

| Label | M7 (streamflow)                                                            | M8 (interaction)                                          |
|-------|----------------------------------------------------------------------------|-----------------------------------------------------------|
| a     | M5 + <code>fatal × streamflow</code> product                               | M5 + <code>fatal × flood_dur</code> (local×local)         |
| b     | M5 with continuous stream <i>replaced</i> by <code>I(stream&gt;p75)</code> | M5 + <code>W.fatal × flood_dur</code> (spillover×local)   |
| c     | M5 + <code>fatal × I(stream&gt;p75)</code>                                 | M5 + <code>fatal × W.flood_dur</code> (local×W-spillover) |
| d     | M5 + <code>I(stream&gt;p75)</code> <i>alongside</i> continuous             | M5 + all three jointly                                    |

**M7 — streamflow upper-tail threshold.** The RF's step-function PDP for streamflow suggests the displacement effect is concentrated in the upper tail. The question is: what functional form best captures this?

`streamflow_anom_1m` is a **signed anomaly** — departures from the climatological mean, with 70% of observations negative (below-average river flow). Two specifications come to mind:

- **Binary dummy** `I(stream > p75)`: 1 when above the 75th percentile (≈ above-average flow), 0 otherwise.
- **Hockey stick** `max(stream − p75, 0)`: 0 below the threshold, excess above it otherwise.

The hockey stick is **not appropriate here**. It implicitly assumes all below-threshold observations contribute exactly zero to displacement — defensible for a non-negative variable (zero rain = zero flood) but unjustified for a signed anomaly where below-average river flow may itself have real effects. Setting 70% of observations to exactly zero discards valid information. The binary dummy avoids this: every observation receives a 0 or 1 without assuming the sub-threshold group has zero effect.

We test the binary dummy in two configurations — replacing the continuous term (M7b) and alongside it (M7d):

| Spec                                          | R²_within     | ΔR²            | <code>I(stream&gt;p75)</code> | continuous stream |
|-----------------------------------------------|---------------|----------------|-------------------------------|-------------------|
| M5                                            | 0.0992        | —              | —                             | +0.0001**         |
| M7b: dummy <i>replaces</i> continuous         | 0.0985        | −0.0007        | +0.115**                      | —                 |
| <b>M7d: dummy <i>alongside</i> continuous</b> | <b>0.1004</b> | <b>+0.0013</b> | <b>+0.099*</b>                | <b>ns</b>         |

M7b is worse than M5 — replacing the continuous term loses information about the full distribution. M7d adds +0.001 R²: the threshold captures the upper-tail jump (+0.099\*), while the continuous streamflow drops to ns. The mechanism is a **directional threshold crossing** — above-average river flow drives displacement; below-average or moderate flow does not.

Critically, **all other M5 coefficients are essentially unchanged in M7d:**

| Driver                       | M5       | M7d      |
|------------------------------|----------|----------|
| <code>fatalities_7_9m</code> | +0.046** | +0.046** |

| Driver              | M5        | M7d       |
|---------------------|-----------|-----------|
| W-fatalities_10_12m | +0.252*** | +0.250*** |
| spi6_13_18m         | +0.139*** | +0.140*** |
| spi6_flood_dur      | −0.054*** | −0.053*** |
| p_anom              | −0.002**  | −0.003**  |

M5 retains the continuous term for parsimony — the threshold adds only +0.001 R<sup>2</sup> and weakens (but does not eliminate) the continuous term. The overall conclusion is unchanged.

**M8 — conflict × flood-duration interaction.** Three product terms are added jointly (local×local **fatal×flood\_dur**, W-conflict×local, local×W-flood\_dur):

| Spec                   | R <sup>2</sup> _within | ΔR <sup>2</sup> | fatal×flood_dur | W-fat×flood_dur | fat×W-flood_dur |
|------------------------|------------------------|-----------------|-----------------|-----------------|-----------------|
| M5                     | 0.0992                 | —               | —               | —               | —               |
| M8a: LL alone          | 0.0992                 | +0.000          | ns              | —               | —               |
| M8d: all three jointly | 0.1017                 | +0.0026         | ns              | ns              | ns              |

**All three interaction terms are insignificant.** The ΔR<sup>2</sup>=+0.003 in M8d reflects the joint model absorbing some shared variance, not any individually significant interaction. Importantly, the M5 linear coefficients remain stable in M8d — conflict and SPI change by less than 10%:

| Driver              | M5        | M8d       |
|---------------------|-----------|-----------|
| fatalities_7_9m     | +0.046**  | +0.052*** |
| W-fatalities_10_12m | +0.252*** | +0.256*** |
| spi6_13_18m         | +0.139*** | +0.138*** |
| spi6_flood_dur      | −0.054*** | −0.059*** |

**M5's additive linear specification holds.** Adding the threshold (M7d) or interactions (M8d) leaves the substantive M5 findings — conflict dominance, SPI wave-absorption, acute water channels — essentially unchanged.

**Summary: RF stress-tested M5 and mostly validated it.** One real threshold (streamflow), one weak interaction (conflict × flood-duration), several artefacts dismissed.

Part III — What M5 averages away: heterogeneity

M5 estimates a single coefficient for each driver — an *average* effect across all 44 cercles and all years. But averages can mask heterogeneity: some cercles may respond strongly to food shocks while others show no response, and the two effects cancel in the average. Part III tests whether cross-sectional moderators (a cercle's displacement history, population size, border location, urban status) reveal structure that M5's averages hide.

**Model numbering note.** Models M11–M17 are all extensions of M5, each adding one interaction or control term. They are presented in logical order — substitution channels (M11, M12), then buffering channels (M13, M14), then the named-disaster test (M15, M16) — and end with the combined model (M17).

The general formula for each interaction test is:

$$y_{it} = \underbrace{[\text{M5 terms}]}_{11 \text{ variables}} + \underbrace{\gamma f(x_{it}) \cdot z_i}_{\text{interaction: driver} \times \text{moderator}} + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

where  $x_{it}$  is a time-varying M5 driver and  $z_i$  is a time-invariant cercle-level moderator (standardised to mean 0, SD 1). Because  $z_i$  is absorbed by the entity FE  $\alpha_i$  when entered alone, only the *interaction* term  $f(x) \cdot z$  varies within a cercle over time and is identified.

M11 — displacement history moderates food and conflict

**Motivation.** M5 shows food is null on average. But maybe food matters in *some* cercles and not others — and the two effects average to zero. A cercle's cumulative displacement history is a natural candidate moderator: where large numbers have already been displaced, a new food shock may translate into displacement differently than in a cercle with little prior displacement.

**Moderators.** All four are constructed the same way: take the **per-cercle time-average** of the transformed variable across all observation periods, then standardise to mean 0, SD 1 across cercles. The time-average is necessary because the underlying variable varies over time within each cercle (IDP stock, food prices, conflict and climate all change year to year). Averaging collapses that time variation into a single stable characteristic of the cercle. A time-invariant moderator is also required for identification: the entity FE absorbs any pure cross-sectional variable, so only the *interaction* (driver × baseline) is identified — it varies over time because the driver varies, while the baseline stays fixed.

| Baseline                 | constructed from                                | interpretation                              |
|--------------------------|-------------------------------------------------|---------------------------------------------|
| <b>food_baseline</b>     | time-avg of f(inflation_food_19_24m) per cercle | persistently high/low food price inflation  |
| <b>conflict_baseline</b> | time-avg of f(fatalities_7_9m) per cercle       | persistently high/low local violence        |
| <b>climate_baseline</b>  | time-avg of f(spi6_13_18m) per cercle           | persistently wet/dry flood exposure history |

| Baseline            | constructed from                    | interpretation                                                       |
|---------------------|-------------------------------------|----------------------------------------------------------------------|
| <b>idp_baseline</b> | time-avg of log1p_idp_t1 per cercle | persistently large/small IDP stock — cumulative displacement history |

#### M11a–d each add one food × baseline interaction:

We test four candidate baselines for the food channel (M11a–d), then extend the winning baseline to the conflict channel:

| Spec        | interaction                | coef            | p                | ΔR <sup>2</sup> |
|-------------|----------------------------|-----------------|------------------|-----------------|
| M11a        | food × food_baseline       | −0.00125        | 0.464 ns         | −0.001          |
| M11b        | food × conflict_baseline   | −0.00248        | 0.061 *          | +0.001          |
| <b>M11c</b> | <b>food × idp_baseline</b> | <b>−0.00541</b> | <b>0.001 ***</b> | <b>+0.003</b>   |
| M11d        | food × climate_baseline    | −0.00048        | 0.728 ns         | +0.001          |

**idp\_baseline** is the clear winner. M11a is null — a cercle's own persistent food price level does not moderate how it responds to food shocks. M11d is also null — persistent flood exposure history (climate\_baseline, constructed as the per-cercle time-average of spi6\_13\_18m) does not moderate the food response either. M11b is marginal (\*) because conflict and displacement history are correlated; idp\_baseline captures the relevant dimension more directly.

**M11c (coef=−0.00541\*, ΔR<sup>2</sup>=+0.003)\*\*** shows that the food null decomposes: in low-IDP-history cercles food stress increases displacement; in high-IDP-history cercles it reduces it. The average is near zero — M5's null — but the heterogeneity is real. The null results for M11a and M11d confirm that it is specifically the *displacement history* of the cercle that matters, not its food price or climate history. What the dampening *means* — whether the food-stressed population is no longer present, or is diverted to a different channel — cannot be settled with IDP data alone. Part IV resolves it directly using the independent refugee outcome: the effect is **cross-border substitution** (food shocks redirect people from internal registration to refugee flight), not a reduction in displacement.

**Having established that displacement history dampens the food response, we ask whether it also dampens the conflict-arrival channel.** The W-fatalities term captures *arrivals* from neighbouring cercles — a cercle that has already absorbed many prior arrivals may have less remaining capacity (housing, services, social networks) to receive further inflows:

| Spec  | interaction          | coef     | p         | ΔR <sup>2</sup> |
|-------|----------------------|----------|-----------|-----------------|
| M11_W | W-fat × idp_baseline | −0.16494 | 0.001 *** | +0.007          |

The conflict-arrival channel is dampened too, and more strongly than food (coef=−0.165\*\*\*, ΔR<sup>2</sup>=+0.007 vs +0.003). Displacement history thus moderates both channels — though the *mechanism* differs: for food it is cross-border substitution (Part IV), while for conflict arrivals the most natural reading is capacity absorption, which the refugee data does not test.

**Caveat.** This conflict interaction is the one finding that does **not** survive the dynamic-panel (Nickell) bias correction (it collapses to ns) — unsurprisingly, since it is built on **idp\_baseline**, which is derived from the lagged displacement level. It should be treated as suggestive only; the food result (M11c) does not share this fragility and is corroborated by the refugee mirror. See *Robustness — dynamic-panel bias*.

#### M12 — border cercles leak food-stressed populations

**Motivation.** Cercles on Mali's borders share a land crossing with neighbouring countries. A food-stressed population near a border has an exit option — cross-border refugee flight — that an interior cercle does not. The food effect may therefore be smaller (or even opposite) in border cercles.

**Moderator.** **is\_border** = 1 for cercles with cumulative refugee outflow > 1,000 (20 cercles identified from UNHCR data).

| Interaction                     | coef            | p              | ΔR <sup>2</sup> |
|---------------------------------|-----------------|----------------|-----------------|
| <b>food × is_border</b>         | <b>−0.00478</b> | <b>0.034**</b> | <b>+0.002</b>   |
| spi6_13_18m × is_border         | +0.039          | 0.365 ns       | −0.001          |
| fatalities_7_9m × is_border     | −0.019          | 0.685 ns       | 0.000           |
| W-fatalities_10_12m × is_border | −0.073          | 0.290 ns       | +0.001          |

Only the food channel is dampened in border cercles. This is **not wave absorption** — it is **leakage**: food-stressed populations near the border substitute cross-border refugee flight for internal IDP registration. The dampened IDP response is not a reduction in total displacement but a reallocation to the refugee channel. This interpretation is confirmed directly in Part IV.

**Sensitivity to border threshold.** **is\_border** is defined as cumulative refugee outflow > 1,000, giving 20 border cercles. Re-running with thresholds of 500 (31 cercles), 2,000 (16 cercles), and 5,000 (11 cercles) gives significant **food × is\_border** at \*\* or \*\*\* in all cases, with coefficients ranging from −0.005 to −0.008. The finding is not sensitive to where the threshold is drawn.

#### M13 — population size buffers climate and food (not conflict)

**Motivation.** Populous cercles (cities, regional capitals) have more diversified economies and deeper food markets. They may absorb livelihood shocks that would displace people in smaller, more food-dependent cercles.

**Moderator.** **log(population\_2015)**, standardised across cercles.

| Interaction                    | coef            | p               | $\Delta R^2$  |
|--------------------------------|-----------------|-----------------|---------------|
| <b>p_anom × log(pop)</b>       | <b>-0.00181</b> | <b>0.010***</b> | <b>+0.003</b> |
| food × log(pop)                | -0.00184        | 0.056*          | +0.001        |
| fatalities_7_9m × log(pop)     | -0.015          | 0.463 ns        | 0.000         |
| W-fatalities_10_12m × log(pop) | -0.037          | 0.327 ns        | 0.000         |

The interaction coefficients tell a precise story about how population moderates each channel.

**Acute rainfall (p\_anom × pop, -0.00181\*)** 🤔\* In small cercles (-1 SD population), the rainfall slope is **+0.00075** — rainfall *increases* displacement. In large cercles (+1 SD), the slope is **-0.00287** — rainfall is strongly protective. The sign actually *flips* across the population distribution. Small, rural cercles are so dependent on rain-fed agriculture that a rainfall anomaly directly affects livelihoods and can push people to move. Populous cercles have diversified non-agricultural income; rainfall variation doesn't threaten livelihoods the same way.

**Food prices (food × pop, -0.00184):**\* The slope stays positive at both ends — food stress increases displacement everywhere — but the magnitude halves: +0.00731 in small cercles vs +0.00363 in large cercles. Populous cercles have deeper markets, more wage employment, and greater ability to substitute away from food price shocks. They don't eliminate the food-displacement relationship; they dampen it by roughly half.

**Conflict (both null):** Violence displaces uniformly regardless of city or cercle size — there is no economic buffer against security threats.

The results are fully robust to population year: re-running with 2019 population gives identical coefficients and p-values to four decimal places, as the cross-cercle population ranking is perfectly stable between 2015 and 2019 (corr = 1.000).

#### M14 — cities buffer climate and food (not conflict)

**Motivation.** Major cities may differ from border or high-population cercles through institutional capacity — market integration, social services, employment diversity. A discrete city dummy tests whether urban status buffers shocks over and above the continuous population effect in M13.

**Moderator.** **is\_urban** = 1 for Bamako and 5 regional capitals (Kayes, Ségou, Sikasso, Mopti, Gao) — 6 cercles.

| Interaction                    | coef            | p               | $\Delta R^2$ |
|--------------------------------|-----------------|-----------------|--------------|
| streamflow × is_urban          | +0.000          | 0.970 ns        | 0.000        |
| <b>spi6_13_18m × is_urban</b>  | <b>-0.087</b>   | <b>0.040 **</b> | -0.003       |
| spi6_flood_dur × is_urban      | -0.033          | 0.328 ns        | -0.003       |
| p_anom × is_urban              | -0.002          | 0.227 ns        | +0.002       |
| <b>food × is_urban</b>         | <b>-0.00798</b> | <b>0.028 **</b> | -0.001       |
| fatalities_7_9m × is_urban     | -0.030          | 0.248 ns        | 0.000        |
| W-fatalities_10_12m × is_urban | -0.034          | 0.586 ns        | 0.000        |

Both food and the slow SPI channel are significantly buffered in cities (both \*\*). The conflict channel is null across all M5 drivers. Cities dampen livelihood-driven displacement through market depth and employment alternatives; they do not change the violence-displacement relationship.

**On the negative  $\Delta R^2$ .** Both significant interactions show  $\Delta R^2 < 0$  —  $R^2_{\text{within}}$  *decreases* when the interaction is added. This is paradoxical only if  $R^2$  is interpreted as a simple goodness-of-fit measure. In a two-way FE panel, **rsquared\_within** is the fraction of within-unit variance explained *after* demeaning. Adding a new regressor can mechanically reduce this metric if the new term absorbs variance that the entity FE was previously accounting for through the demeaned mean — a known property of PanelOLS, not a sign that the term is harmful. The interaction coefficient being significant ( $p < 0.05$ ) and mechanistically coherent is the correct criterion; the negative  $\Delta R^2$  is a measurement artefact of the FE framework, not evidence against the finding.

**Sensitivity to urban definition.** We test eight alternative definitions to identify the mechanism behind M14's findings:

| Definition                  | n        | Mopti/Gao? | food×urban      | spi6×urban      | W-fat×urban |
|-----------------------------|----------|------------|-----------------|-----------------|-------------|
| Bamako + Mopti + Gao        | 3        | ✓          | -0.010**        | ns              | ns          |
| Kayes + Ségou + Sikasso     | 3        | ×          | ns              | ns              | ns          |
| <b>All 6 named capitals</b> | <b>6</b> | <b>✓</b>   | <b>-0.008**</b> | <b>-0.087**</b> | <b>ns</b>   |
| Top-6 by population         | 6        | ×          | ns              | ns              | ns          |
| Top-10 by population        | 10       | ✓          | ns              | ns              | ns          |
| Top-6 by density            | 6        | ×          | ns              | ns              | ns          |
| Top-10 by density           | 10       | ✓          | ns              | ns              | ns          |

Four conclusions emerge. First, **W-fatalities is null across every single definition** — conflict displaces uniformly regardless of how urban status is defined. This is the most robust result in M14: there is no urban buffer against violence under any reasonable definition.

Second, **Kayes, Ségou and Sikasso — despite being larger (population ranks 3–5) and denser than Mopti and Gao — contribute nothing alone**, ruling out population size and density as the mechanism for the food and SPI effects.

Third, **simply including Mopti and Gao in a larger set is not sufficient**: top-10 definitions that include both produce null results, because diluting them with 7–8 non-capital cercles removes the signal.

Fourth, the food and SPI effects are specific to the **6-capital administrative network** as a coherent system. Gao (density rank 34, population rank 12) and Mopti (population rank 10) are not large or dense cities. What they share with Bamako is their role as **regional administrative and commercial hubs** for Mali's most food-insecure and climate-exposed populations. The buffering reflects institutional access — markets, government services, NGO presence — not demographic scale.

M15/M16 — Do named disasters amplify the climate-displacement response?

**Motivation.** The core question is whether officially declared flood or drought events amplify the displacement response to climate variables — does the SPI or streamflow slope steepen during a named disaster period? If so, the binary event label captures severity information that the continuous physical variables miss.

**Data.** 20 EMDAT flood + drought events in Mali 2007–2024, spatially assigned to cercles by admin level (admin2 directly; admin1 fanned out to all cercles in the region; admin0 nationally). **flood\_active** = 1 for any observation interval overlapping an active flood event for that cercle.

Before testing interactions, we first verify that **flood\_active** carries any independent displacement signal at all (M15). If the dummy is null as a main effect, there is little motivation to test whether it amplifies slopes.

**M15** — M5 plus event dummy as main effect:

$$y_{it} = [M5] + \gamma \cdot \text{flood\_active}_{it} + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

| Spec                      | coef          | p              | R <sup>2</sup> _within | ΔR <sup>2</sup> |
|---------------------------|---------------|----------------|------------------------|-----------------|
| M5 baseline               | —             | —              | 0.0992                 | —               |
| M15a: M5 + flood_active   | <b>+0.153</b> | <b>0.038**</b> | 0.0983                 | −0.001          |
| M15b: M5 + drought_active | −0.015        | 0.788 ns       | 0.0990                 | 0.000           |

**flood\_active** is significant (+0.153\*, p=0.038): named floods carry a displacement signal beyond the continuous physical intensity. **drought\_active** is null — the continuous channels fully encode drought severity. The negative ΔR<sup>2</sup> for M15a is the FE non-monotonicity artefact. All M5 coefficients remain stable (W-fatilities changes by < 1%, SPI channels by < 2%). This confirms named flood events are worth testing as slope amplifiers.

**M16** — does the named event steepen the climate slope? We add an interaction between each continuous climate variable and the event dummy:

$$y_{it} = [M5] + \gamma \cdot \text{flood\_active}_{it} + \delta \cdot (f(x_{it}) \times \text{flood\_active}_{it}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

| Spec              | interaction            | coef     | p              |
|-------------------|------------------------|----------|----------------|
| M16a              | streamflow × flood     | +0.00012 | 0.277 ns       |
| M16b              | p_anom × flood         | +0.00055 | 0.756 ns       |
| M16c              | spi6 × flood           | +0.113   | 0.074 *        |
| M16d              | spi6_flood_dur × flood | +0.063   | 0.086 *        |
| M16e              | spi6 × drought         | −0.035   | 0.781 ns       |
| M16f              | p_anom × drought       | −0.006   | 0.100 ns       |
| <b>M16g joint</b> | <b>all 6 together</b>  | —        | <b>0/6 sig</b> |

M16c and M16d are marginally significant in isolation, but **0/6 survive the joint test**. The amplifier hypothesis fails: during named flood events, the displacement slope on SPI and streamflow does not systematically steepen. The continuous variables capture the relevant physical intensity; the named-event label adds a level shift (M15) but not a slope change (M16).

M17 — unified stacked model

M7–M16 each tested one extension in isolation. M17 stacks the four validated extensions into one specification to check whether they survive jointly and to measure their combined explanatory power.

**The four extensions in M17** — each from a validated standalone test:

- **food × idp\_baseline (M11c)** — food dampened in high-displacement cercles (cross-border substitution; Part IV)
- **W·fat × idp\_baseline (M11\_W)** — conflict spillover also dampened
- **I(streamflow > p75) (M7d)** — current-window flood threshold (added alongside the continuous streamflow already in M5)
- **p\_anom × log(pop) (M13)** — acute rainfall buffered in populous cercles

$$y_{it} = [M5] + \gamma_1(f(\text{food}) \cdot z_{\text{idp}}) + \gamma_2(f(W \cdot \text{fat}) \cdot z_{\text{idp}}) + \gamma_3 \cdot \mathbf{1}[\text{stream} > p_{75}] + \gamma_4(f(p\_anom) \cdot z_{\text{pop}}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

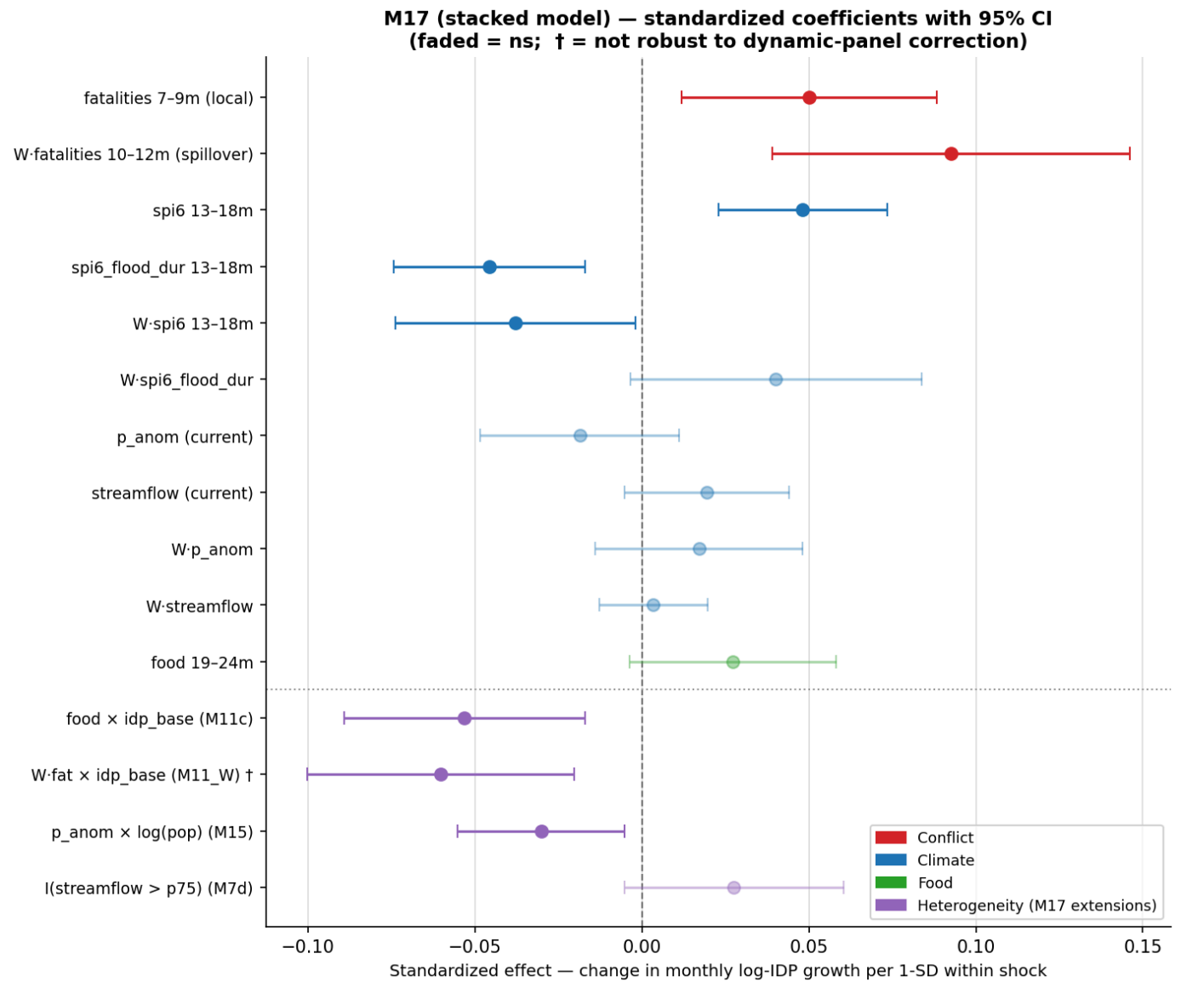
**Results:**

| Term             | coef     | p     |     |
|------------------|----------|-------|-----|
| food × idp_base  | −0.00501 | 0.004 | *** |
| W·fat × idp_base | −0.14461 | 0.003 | *** |



| Term            | coef     | p     |    |
|-----------------|----------|-------|----|
| p_anom × pop    | −0.00161 | 0.018 | ** |
| I(stream > p75) | +0.08526 | 0.101 | ns |

**Three of the four survive jointly.** The streamflow threshold, marginally significant standalone (M7d, +0.099\*), weakens to ns when stacked — its upper-tail signal overlaps with the other terms once all are in the model. The three heterogeneity interactions (M11c, M11\_W, M13) remain strongly significant. R<sup>2</sup><sub>within</sub> rises from **0.099 to 0.112** (+0.013).



**Figure 6.** M17 (stacked model) coefficients as **standardized effects** (change in monthly log-IDP growth per 1-SD within-circle shock), with 95% CIs; faded = not significant. Above the line are the M5 main effects; below are the four heterogeneity extensions (M11c, M11\_W, M15, M7d threshold). The acute-water main effects (*p\_anom*, *streamflow*) lose *individual* significance here because their signal is partly absorbed by the *p\_anom × log(pop)* interaction and the streamflow threshold. † *W-fat × idp\_base* does not survive the dynamic-panel correction (see *Robustness*).

A notable side effect: the *W-fatalities* main effect **strengthens from +0.252 to +0.373** in M17. M5's coefficient was an average across the IDP-history distribution; M17 separates the absorption component (the *W-fat × idp\_baseline* interaction) from the baseline conflict effect, so the main coefficient now represents the conflict effect in *low*-IDP-history circles where this dampening is minimal.

**Model progression.** Before decomposing M17, here is how far the modelling has come:

| Model                 | R <sup>2</sup> <sub>within</sub> | Δ      |
|-----------------------|----------------------------------|--------|
| Controls + FE only    | 0.048                            | —      |
| M5 (selected drivers) | 0.099                            | +0.051 |
| M17 (+ heterogeneity) | 0.112                            | +0.013 |

The Stage A–E funnel more than doubled within-R<sup>2</sup> over the controls-only baseline (+0.051); the RF-guided functional-form tests and heterogeneity extensions (M6–M14) added a further +0.013. The LOO below shows which families drive M17's R<sup>2</sup>=0.112.

**LOO block decomposition (comparable basis).** Each block's share of its model's driver contribution — the  $R^2_{\text{within}}$  it gains over the AR-control baseline (0.048) — placed side by side with the M5 decomposition:

| Driver block                                                                          | M5         | M17        |
|---------------------------------------------------------------------------------------|------------|------------|
| <b>Conflict</b> (local + spillover; + $W \cdot \text{fat} \times \text{idp}$ in M17)  | <b>72%</b> | <b>83%</b> |
| Climate (8 water/SPI terms; + threshold & $p_{\text{anom}} \times \text{pop}$ in M17) | 16%        | 8%         |
| Food (+ $\text{food} \times \text{idp}$ in M17)                                       | ~0% (null) | 6%         |

Conflict dominates both models — its share *rises* from 72% to 83% once the  $W \cdot \text{fat} \times \text{idp\_baseline}$  interaction joins the conflict block. Food climbs from null to 6%, the  $\text{food} \times \text{idp\_baseline}$  interaction unlocking signal M5 was averaging away. Within the climate block, the slow SPI terms contribute essentially nothing once stacked; the acute water terms carry it.

Temporal stability — the 2020 structural break

Mali's panel spans a major political rupture: the **military coup of 18 August 2020**, after which the international counter-insurgency presence wound down and the conflict shifted toward a decentralised rural insurgency. Splitting the sample at 2020-07 (just before the coup) reveals that the displacement regime does not merely weaken — it **inverts**.

| M5 channel                                        | Pre-2020-07 | Post-2020-07                                           |
|---------------------------------------------------|-------------|--------------------------------------------------------|
| n                                                 | 1,111       | 306                                                    |
| $R^2_{\text{within}}$                             | 0.104       | <b>0.365</b>                                           |
| local conflict (fat 7–9m)                         | +0.051 ns   | <b>+0.044 ***</b>                                      |
| conflict spillover ( $W \cdot \text{fat}$ 10–12m) | +0.264 **   | <b>–0.110 ** (collapses; ns after bias correction)</b> |
| SPI flood (spi6 13–18m)                           | +0.150 ***  | +0.044 ns                                              |
| flood duration                                    | –0.065 ***  | –0.020 ns                                              |
| current rainfall ( $p_{\text{anom}}$ )            | –0.0026 **  | +0.0002 ns                                             |
| streamflow                                        | +0.00011 ** | –0.00001 ns                                            |
| food (19–24m)                                     | +0.014 **   | +0.001 ns                                              |

Three things change at once. First, **the model fits far better post-coup** ( $R^2_{\text{within}}$  0.10 → 0.37): displacement became much more predictable, because a single factor came to dominate. Second, **that factor is local conflict** — *fatalities\_7\_9m*, non-significant before, becomes the strongest driver after. Third, and most striking, **the conflict spillover collapses** — the robustly positive pre-2020 effect (+0.26) vanishes, turning to –0.11 in FE (and to a non-significant –0.06 under the dynamic-panel correction; see *Robustness*). Violence in neighbouring cercles no longer pulls people in as IDP arrivals.

The coherent explanation is that **post-coup road blockages broke the spatial flight mechanism**. Before 2020, people fled conflict by moving toward safer neighbouring cercles, registering as arrivals there — the positive spillover. After 2020, with roads cut and movement restricted, violence displaces people *in place* and cross-cercle flight shuts off — which is exactly why the spillover channel **vanishes** (a switched-off reception channel implies a *null* spillover; the FE estimate even turns negative, but the bias-corrected estimate is indistinguishable from zero). Every climate and food channel — which operate on slower, livelihood-mediated timescales — is swamped by the violence and goes null.

**Is 2020 really the break?** Re-splitting at every half-year from 2018 to 2021 confirms the break is genuinely around the coup, not an artefact of the chosen cut:

| Split point    | n_post | $R^2_{\text{post}}$ | spillover (post) |
|----------------|--------|---------------------|------------------|
| 2018-07        | 620    | 0.15                | –0.049 ns        |
| 2019-07        | 445    | 0.19                | –0.090 ns        |
| <b>2020-07</b> | 306    | <b>0.37</b>         | <b>–0.110 **</b> |
| 2021-07        | 219    | 0.42                | –0.143 **        |

Two signatures pin the break to 2020. First, the post-period  $R^2$  jumps sharply — about 0.15 for pre-2020 cuts, but 0.31–0.42 once the cut reaches 2020. Second, the collapse of the positive spillover is only *significant* from 2020 onward: cutting at 2018 or 2019 gives a weak, non-significant negative spillover (those post-windows are still mostly pre-coup), but as the cut moves into the post-coup period the negative spillover strengthens monotonically (–0.09\*\* → –0.14\*\*\*, in FE). The structural break is a 2020 phenomenon.

**All magnitudes in this report describe 2014–2020 Mali.** The post-2020 regime is a different system, and the static road-distance spatial weights cannot represent blocked roads; forecasting the current period would require a road-state-aware weight matrix and re-estimation on a recent subsample.

Part IV — External validation: refugee outflow

DTM IDP data and UNHCR refugee data are collected by different agencies, using different methodologies, recording different physical outcomes. If the same M5 drivers predict refugee outflow with the same signs, that is strong cross-outcome validation.

The refugee outcome is each cercle's outflow of Malian refugees **across the border to neighbouring countries**. Because this is movement *out of Mali*, the spatial-spillover terms — which capture redistribution *between* Malian cercles — do not apply: cross-border flight is driven by the cercle's *own* conditions

pushing people over the international border, not by neighbour-weighted conditions inside Mali. The refugee model therefore uses local drivers only.

**The refugee data is sparse and geographically concentrated.** Unlike internal displacement, which is spread across the country, cross-border outflow is overwhelmingly a **northern** phenomenon: ~80% originates in the three northern regions (Tombouctou 41%, Taoudénit 23%, Gao 15%), and just **three cercles account for ~half** of all outflow (the top six ≈ 74%). It traces to the post-2012 occupation of the north, with refugees fleeing chiefly to **Mauritania, Niger, and Burkina Faso**, and the series runs longer than the IDP panel (2012–2025). The refugee model is therefore identified off a smaller, northern base than the broad-coverage IDP model — which makes the cross-outcome agreement below (especially the mirror substitution) a check across genuinely *different* geographies, but also means the refugee findings lean on a handful of northern cercles and should be read as corroboration rather than independent proof.

**The refugee outcome is well explained by the same drivers ( $R^2_{\text{within}} = 0.119$ ), but with a different driver mix.** Comparing each driver on both outcomes under a common local specification (recent conflict at 1–3m):

| Driver                    | refugee     | IDP        |
|---------------------------|-------------|------------|
| local conflict (1–3m)     | –0.016 ns   | +0.039 *   |
| SPI flood (13–18m)        | +0.175 *    | +0.115 *** |
| flood duration (13–18m)   | –0.059 ns   | –0.047 *** |
| current rainfall (p_anom) | +0.0052 *** | –0.0020 ** |
| streamflow                | +0.000 ns   | +0.0001 ** |
| food (19–24m)             | +0.016 *    | +0.0065 *  |

Two contrasts stand out. **Refugee outflow is driven by climate and food, not conflict** — local conflict is non-significant for refugees, whereas it predicts IDP (and far more strongly through neighbour *spillover*: the M5 structure reaches 0.099 vs 0.058 for this local spec). And the climate channels cross-validate by sign: **SPI flood carries the same positive sign on both outcomes** (+0.175\* refugee, +0.115\*\*\* IDP), while **current rainfall flips sign** (+0.0052\*\*\* refugee vs –0.0020\*\* IDP) — protective for inland livelihoods (less IDP) but causing Niger River flooding that drives refugees northward. Food carries the same positive sign on both.

**Why food appears to dampen displacement.** Part III left a puzzle: food-price shocks seemed to *reduce* internal displacement in cercles with a long displacement history (M11c) and those near a border (M12). Why would food stress reduce flight? The refugee data answers it. The same two interactions, run on refugee outflow, come out **equal and opposite** to the IDP versions:

| Interaction                | refugee    | IDP        | ratio        |
|----------------------------|------------|------------|--------------|
| food × idp_baseline (M11c) | +0.0057*** | –0.0054*** | <b>–1.05</b> |
| food × is_border (M12)     | +0.0080**  | –0.0050**  | <b>–1.61</b> |
| p_anom × log(pop) (M13)    | ns         | –0.0016**  | IDP-specific |
| food × is_urban (M14)      | ns         | –0.0081**  | IDP-specific |

In these cercles a food-price shock does not dampen displacement — it **redirects** it, shifting people from internal IDP registration to cross-border refugee flight (near 1:1 for displacement-history cercles, larger still for border cercles). Food stress is a *substitution between displacement channels*, not a reduction in displacement. The other two interactions (M13, M14) have no refugee counterpart — they are genuinely IDP-specific.

### Robustness — dynamic-panel (Nickell) bias

M5 conditions on the lagged displacement level (`log1p_idp_t1`) to absorb mean reversion. In a two-way fixed-effects panel, including a lagged dependent level induces **Nickell bias**, which can distort the driver coefficients. We test this with an instrumental-variables correction.

**The instrument.** The natural Anderson–Hsiao instrument is the previous survey's level, but it is *invalid here*: because the outcome is a **change** over consecutive intervals that share an endpoint, the residuals carry negative serial correlation ( $AR(1) \approx -0.23$ ). We therefore use the **lag-2 level** as the instrument (the Arellano–Bond remedy under MA(1) errors); its first stage is strong.

| Quantity                        | FE               | IV (lag-2)       | verdict                  |
|---------------------------------|------------------|------------------|--------------------------|
| M5 conflict spillover           | +0.251***        | +0.208***        | robust (–17%)            |
| food × idp_base (M11c)          | –0.0050***       | –0.0030**        | robust                   |
| p_anom × log(pop) (M15)         | –0.0016**        | –0.0018***       | robust                   |
| l(stream > p75)                 | +0.085 ns        | +0.080 ns        | stable                   |
| <b>W-fat × idp_base (M11_W)</b> | <b>–0.145***</b> | <b>–0.020 ns</b> | <b>not robust</b>        |
| spillover, pre-2020             | +0.264**         | +0.241***        | robust                   |
| spillover, post-2020            | –0.110**         | –0.062 ns        | sign holds, sig. weakens |

Three conclusions:

1. **The headline results are robust.** The conflict spillover (M5), the food-substitution interaction (M11c), and the population buffer (M15) all survive the correction with the same signs and significance; the bias shaves only ~17% off the conflict spillover.

2. **The structural break holds in direction.** The spillover is positive pre-2020 (+0.241\*\*\*) and negative post-2020 (−0.062); the reversal survives the correction, though the post-coup estimate loses significance in the smaller IV sample (n≈300).
3. **One exception — the conflict wave-absorption interaction ( $W \cdot fat \times idp\_base$ ,  $M11\_W$ ) is not robust:** it collapses to non-significance. This is expected — it is built on  $idp\_baseline$ , which is mechanically derived from the lagged level, making it the term most entangled with the dynamic-panel bias. The *food* substitution result (M11c) does not share this fragility and is independently corroborated by the refugee mirror (Part IV).

Conclusion — what we found

What drives displacement:

- **Violence is the biggest driver — and it spreads.** Violence in *neighbouring* communities pushes people to flee even more than violence at home — about five times more.
- **Climate matters, but less — and both extremes push people out.** Severe flooding about a year earlier, and sudden rainfall or river surges in the moment, drive displacement — and so does drought, on a longer horizon (about two years). All are secondary to violence, but real.
- **Food prices don't push people out — they redirect them.** This shows up most clearly in two kinds of places: communities with a long displacement history, and **those on a border with a ready cross-border exit**. In both, food-price spikes send people **across the border as refugees** rather than into internal displacement — the same number flee, but through a different channel.

Who is affected depends on local conditions:

- The same shock lands differently depending on a community's history and size. Places with a long displacement history, and larger towns and cities, **absorb or redirect** food and climate shocks — but **not violence, which displaces people everywhere regardless**.
- **Yet when we combine every driver and every one of these differences into a single model, the verdict is unchanged: conflict alone accounts for about half of everything we can explain; climate and food make up the smaller remainder.**

How much this explains, and when:

- The machine-learning cross-check confirmed the findings and added a sobering note: **displacement is genuinely hard to predict**. Much of why people flee comes down to local factors no dataset captures — community ties, leadership, household decisions.
- An independent refugee dataset — different agency, different outcome — reproduced the climate and food patterns, which strengthens confidence that these are real signals.
- The main results also survive a **dynamic-panel robustness check** (correcting for how the model carries past displacement forward): the conflict, climate, and food-redirection findings all hold. The one exception is a *secondary* pattern — the dampening of conflict spillover in already-displaced areas — which does not survive and should be treated with caution.
- These patterns describe **Mali up to 2020**. After the August 2020 coup the whole pattern shifts: local violence becomes the dominant driver, and the cross-community spillover **collapses** — blocked roads appear to cut off flight to neighbours, so the channel that pulled people into safer cercles switches off (the effect drops from strongly positive to indistinguishable from zero), while the climate and food signals fade. The model explains the past well; the post-coup period is a different system.

The findings in detail:

| Driver                  | finding                                                                                                                           | robustness                                                                            |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Conflict spillover      | Dominant. Violence in neighbouring cercles (10–12m lag) is the single largest predictor. Spillover 5x local.                      | 100% of 987k models; stable across conflict source, spatial weights, SE specification |
| SPI flood channel       | Real but secondary. Wet anomaly 13–18m ago → displacement; prolonged prior flooding → wave absorbed.                              | 272/5,880 both-significant pairs; RF confirms linear, not threshold                   |
| Acute water shock       | Real and contemporaneous. Current rainfall protective; active streamflow displacing.                                              | Threshold confirmed (M7d); robust across lag specs                                    |
| Food prices             | Null on average. Redirects rather than suppresses — food shocks shift displacement from internal (IDP) to cross-border (refugee). | All 21 food variables null in multivariate; M11c/M12 mirror in refugee data ***       |
| Food spillover (W-food) | Not informative. Dropped — p=0.729; food price is a near-national time series.                                                    | Confirmed across all 21 food variables                                                |

Limitations and open questions

**Identification.** TWFE with banded lags and spatial spillovers provides *conditional associations* suggestive of causal effects, not clean causal estimates. No instrument or natural experiment is available **for the drivers themselves** (the dynamic-panel correction above instruments only the lagged outcome, not conflict, climate, or food). The causal interpretation rests on design choices: the conflict lags (local 7–9m, spillover 10–12m) make reverse causality implausible;  $W \cdot fatalities$  is hard to argue is caused by local IDP arrivals; two-way FE removes the dominant confounders. Remaining time-varying cercle-specific confounders are not controlled for.

**R<sup>2</sup> and causal importance.** R<sup>2</sup><sub>within</sub> is the primary fit criterion for model selection, applied alongside the p-value and coefficient-sign criteria. It is imperfect — an *in-sample* measure, and non-monotonic under the two-way FE transform — but it is the natural, interpretable metric in the within-cercle space, and the selections were cross-checked against out-of-sample R<sup>2</sup> (RF cross-validation). Crucially, R<sup>2</sup><sub>within</sub> measures explanatory fit, not causal effect size: the conflict > climate > food ranking reflects relative *explanatory* importance, not a direct measure of causal contribution.

**Incomplete non-linearity scan.** The Random Forest non-linearity screen (Part II) was computed on the M5 feature set, so non-linear structure in non-selected variables — for example a threshold at a lag band outside M5 — is not screened and could be missed. The risk is one of omission only: every candidate still faces a de-confounded TWFE test, and the full-feature RF bounds the missed structure to be small.

**Two-lag SPI.** The slow flood signal (spi6 13–18m) and a longer-horizon drought signal (spi6 19–24m) are each individually significant (drought  $-0.064^{**}$ ,  $p=0.015$ ) and nearly orthogonal in the within-variation (within-correlation 0.03). Adding the drought term alongside the flood term leaves every other M5 coefficient essentially unchanged but *lowers*  $R^2_{\text{within}}$  ( $0.099 \rightarrow 0.094$ ) — the fixed-effects non-monotonicity discussed earlier — so M5 keeps only the flood signal for parsimony. The drought channel is real (it appears in Stage A and is examined in M6) but is not in the final model; see *A second drought channel* in Part II.

**Post-2020 regime change.** All magnitudes in this report describe 2014–2020 Mali. After the August 2020 coup the displacement regime shifts — local conflict becomes dominant and the positive cross-circle spillover collapses (see *Temporal stability — the 2020 structural break*). The spatial weight matrix assumes static road connectivity, which post-coup road blockages violate; forecasting the current period would require a road-state-aware  $W$  and re-estimation on a recent subsample.